CLAYTON BRADEN

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Education

University of Colorado Boulder - Boulder, CO

- Applied Compute Science B.S. December 2026*
  GPA: 3.81

Colorado School of Mines - Golden, CO

- Engineering Technology and Management M.E. May 2020
- Chemical and Biochemical Engineering B.S. May 2019

Experience

Service & Operations Development Engineer
Arcscan Inc., Golden, CO **February 2024-Present**

- **Redesigned the organization's complaint intake site** with Python and JavaScript, implementing an end-to-end custom complaint-approval workflow that improved usability, streamlined operations, and enabled KPI tracking/reporting.
- **Reduced overall data storage intake by 10%** by developing a program that identifies redundant or outdated sales, engineering, and manufacturing data.
- **Developed a self-service product ordering portal** integrated directly with QuickBooks, eliminating the need for third-party purchasing systems, reducing operational overhead, and improving order fulfillment speed by 25%.
- **Integrated REST-based APIs to communicate between various applications**, enabling seamless data flow between manufacturing and engineering teams and improving process-line efficiency.

Manager of Field Service Engineering & Operations
Arcscan Inc., Golden, CO **February 2022-January 2024**

- **Developed and managed service operations department** for medical device startup company focused on robotic ophthalmological imaging.'
- **Reduced total service operations costs by 15%** through improved parts sourcing and manufacturing process'.
- Managed internal business operations by delegating responsibilities and tasks to various departments and individuals.
- Worked directly with suppliers to forecast product failure rates, predicted inventory needs to minimize inventory turnover, and improvements to designs.

Root Cause Analysis Engineer
Arcscan Inc., Golden, CO **May 2020-February 2022**

- **Served as lead to product failure investigation team for medical device company focused** on robotic ophthalmological imaging in addition to providing field service engineering support for product failures and general maintenance both remotely and on-site
- **Determined manufacturing defects and implemented corrective actions**, process changes, and designs directly with contract manufacturers to reduce product failure rates by more than 30%.
- **Designed and executed final product reliability test protocols** resulting in a 20% overall reduction in out-of-box failures.
- **Constructed product manufacturing troubleshooting guide** for improved internal training and faster on-site resolution times.

CS Projects

LitFlix Modern Book Browsing Website | Python, JavaScript

- Designed and built a full-stack web application for browsing books using Project Gutenberg data with 3 other team members.
- Developed a PostgreSQL database with over 1,000 books with structured author, subject, and metadata records.
- Implemented a REST API in Flask to enable efficient communication between the frontend and the database.
- Deployed the application on Render: https://team1-w5mb.onrender.com/home.

2D Bipedal Walker using Deep Reinforcement Learning | Python

- Built a physics-based simulation of a bipedal agent to autonomously learn walking behavior in a 2D simulated environment (OpenAI Gymnasium) via reinforcement learning.
- Designed reward shaping strategies to encourage balance, forward velocity, and smooth gait patterns to improve performance over each episode.
- Modeled continuous state and action space such as joint angles, velocities, and torque outputs.
- Evaluated performance based on episode reward convergence and gait consistency

Pickleball Video Understanding (Early-Stage ML) | Python

- Built and trained a model to interpret pickleball points using only video data, without predefined annotations.
- Trained model using 10-second labeled clips of 4 different matches using Youtube video clips and FFmpeg for extraction.
- Built Python Flask application to easily filter through labeling by watching clip, pressing buttons based on the label, and sending clips to a specific folder for labeling.
- Applied computer vision techniques (CNN's and Kalman filter's) to detect and segment key gameplay actions across time, as well as filter irrelevant frames and transitions.

ArcScan Production Robotics Speed Improvement | C#, .NET API

- Improved medical device robotics scanning speed by 50% by diagnosing and fixing a regression in the latest released production code
- Conducted in-depth analysis of recent GitHub merges to trace performance degradation to a specific code change
- Identified a bug in the logging output that was causing threads of the motion controller to run in reduced parallelism.

Skills & Certifications

Programming Languages

- C++, Python, SQL, Java, JavaScript, Scala -
- C#, .NET API
- Qt Creator with C++ (GUI)

Other Skills

- AutoCAD, Simulink, SolidWorks
- GalilTools (Motion controller application)
- Agile certified